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MILITARY SPECIFICATION

CONTROL SECTION, GUIDED MISSILE

FOR

AIM/RIM-7M, GUIDED MISSILE

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Control Section, Guided Missile, referred to herein as the "assembly".

1.2 Classification. The following control section configurations are covered by this specification.

Type I:	WCU-5/B, Drawing 917AS6200-1 and 2.
Type II:	WCU-5A/B, Drawing 917AS6200-3, WCU-5B/B, Drawing 917AS9074, or WCU-5C/B, Drawing 917AS9868.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 (U) Control section classes. Control sections described by this specification are of the following classes, depending on the level of procurement.

a. Class I: Control sections fabricated for procurement at the missile guidance set level.

b. Class II: Control sections fabricated for procurement at the control section level (see 3.1.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-G-85142	Guidance-Control Sections for Guided Missile AIM/RIM-7M.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

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2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

917AS213	Assembly, Wing and Stud.
917AS216	Stud, Wing.
917AS754	Wing, Guided Missile BSU-56A/B.
917AS6200	Control Section, Guided Missile.
917AS8148	Accumulator and Flow Valve.
917AS9074	Control Section, Guided Missile WCU-5B/B.
917AS9868	Control Section, Guided Missile WCU-5C/B.

RAYTHEON COMPANY
(Code Ident 49956)

479512	Control Section Housing.

(Copies of specifications, standards, drawings, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The assembly shall consist of an autopilot, hydraulic control section, accumulator and flow valve assembly, alternating current (ac) to direct current (dc) converter, craft gyro assembly and missile tune converter.

3.1.1 Class I control sections. Class I control sections shall meet all requirements specified herein.

3.1.2 Class II control sections. Class II control sections shall meet all requirements specified herein and in addition, each control section shall be tested in accordance with and meet all requirements of MIL-G-85142 with the exception that the reliability demonstration tests and contractor tests are not required. The vibration screening and preconditioning and electrical performance tests shall be conducted in accordance with MIL-G-85142 with a mating section.

3.1.3 Government-furnished property list. Unless specified in the contract or purchase order (see 6.2.1), the Government will provide a list of all Government-furnished property to the contractor.

3.2 Characteristics.

3.2.1 Electrical performance. Each assembly shall meet the electrical performance requirements specified herein.

3.2.1.1 AC/DC conversion. The outputs at the following contacts shall be as specified in Table I.

3.2.1.2 Gyro signals.

3.2.1.2.1 Over voltage period. The voltage at J22-15 (1200 Hz phase A) shall be 39.5 ±3.0 Vtrms and at J22-34 (1200 Hz phase B) shall be 39.5 ±3.0 Vtrms when measured in the 2 seconds (s) interval that the 400 Hz power of 4.1.5.2 is applied.

3.2.1.2.2 Normal run period. The output at J22-15 (1200 Hz phase A) shall be 15.0 ±1.5 Vtrms and at J22-34 (1200 Hz phase B) shall be 15.0 ±1.5 Vtrms.

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TABLE I. AC/DC conversion.

Contacts	Vdc 1/	Vtrms 2/
J20-D (Type 1)	+30 ±5	5.5 ±1.5 (400 hertz (Hz))
J20-D (Type II)	+26 ±5	15.0 ±4.5 (400 Hz)
J20-T	+13 ±2	
J20-V	+13 ±2	
J21-13	-26.5 ±2.5 3/	
J21-15	+43.5 ±1.5	
J21-17	+43.5 ±1.5	
J21-18		113.0 ±6.5 (400 Hz)
J21-19		113.0 ±6.5 (400 Hz)
J21-37		113 0 ±6.5 (400 Hz)
J21-32	+26.5 ±2.5 3/	
J21-33	-43.5 ±1.5	
J21-35	-43.5 ±1.5	
J21-27	+53 ±5	
J23-1		113.0 ±6.5 (400 Hz) 3/
J23-2		113.0 ±6.5 (400 Hz) 3/
J23-3		113.0 ±6.5 (400 Hz) 3/
J23-11	-20.6 ±2.0 Vdc 3/	
J23-14	+20.6 ±2.0 Vdc 3/	
J23-15	-20.6 ±2.0 Vdc 3/	
J23-16	+20.6 ±2.0 Vdc 3/	

1/ Volts direct current (Vdc).

2/ Volts true root-mean-square (Vtrms).

3/ Measurements shall be performed within 2 seconds of grounding J20-U.

3.2.1.3 Prelaunch tuning logic. The assembly shall meet the following requirements:

a. No signal. The output at J23-14 shall be +20.6 ±2.0 Vdc for 2.17 ±0.83 s and the output at J20-S shall be not more than +1 Vdc when measured not less than 3 s after J20-U is grounded.

b. No retune. The output at J23-14 shall be 0.0 ±0.5 Vdc.

c. No radar on. The output at J23-14 shall be 0.0 ±0.5 Vdc.

d. Enable command. The output at J23-14 shall be +20.6 ±2.0 Vdc when measured during the interval that the +26.5 volts (V) input to J23-21 is applied.

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3.2.1.3.1 Prelaunch tune. The output at J20-S shall change from 0.0 ±0.5 Vdc to +5.5 ±1.0 Vdc not less than 80 milliseconds (ms) and not more than 400 ms after the input to J23-33 is changed from a ground to +4.5 Vdc. The output at 16J1-10 and J20-f shall be +25.5 ±1.0 Vdc.

3.2.1.3.2 Missile ready latch. The output at J20-S shall be +5.5 ±1.0 Vdc. The output at 16J1-10 shall be 0.0 ±0.5 Vdc.

3.2.1.4 Roll drift. The output at 16J1-21 shall not change at a rate in excess of ±10 millivolts direct current per second (mVdc/s).

NOTE: Adjustment of 16A3R13 is allowed to meet this requirement.

3.2.1.5 Pitch and yaw launch drift/C-band. The outputs at 16J1-19 and 16J1-20 shall not change at a rate in excess of ±35 mVdc/s.

NOTE: Adjustment of 16R1 and 16R2, respectively, are allowed to meet these requirements.

3.2.1.6 Wing servos 1 and 3. The output at 16J1-1 shall lag the 10 Hz input by not greater than 217 degrees (deg).

3.2.1.6.1 Gain margin. The output at 16J1-1 shall be not greater than 0.05 volts root-mean-square (Vrms) when measured not less than 3 s after the pulse to 16J1-14 has been applied.

3.2.1.7 Wing servo 2. The output at 16J1-2 shall lag the 10 Hz input by not greater than 217 deg.

3.2.1.7.1 Gain margin. The output at 16J1-2 shall be not greater than 0.050 Vrms when measured not less than 3 s after the pulse to 16J1-15 has been applied.

3.2.1.8 Wing servo 4. The output at 16J1-3 shall lag the 10 Hz input by not greater than 217 deg.

3.2.1.8.1 Gain margin. The output at 16J1-3 shall be not greater than 0.05 Vrms when measured not less than 3 s after the pulse to 16J1-16 has been applied.

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3.2.1.9 Roll autopilot/3 Hz - 2 degrees peak-to-peak (deg p-p) roll yozzle.

3.2.1.9.1 Residual. The output at 16J1-4 shall be 0 ±50 millivolts direct current (mVdc), at 16J1-24 shall be 117 ±6 mVdc and at 16J1-25 shall be 117 ±6 mVdc.

3.2.1.9.2 Demodulators. The 3 Hz output at 16J1-4 shall be negative for positive rotation (see Figure 1) and shall be 1.18 ±0.12 volts peak-to-peak (Vp-p), and the output at J23-5 shall be 754 ±75 millivolts peak-to-peak (mVp-p). The outputs at 16J1-5 and 16J1-6 shall be not greater than 75.5 mVp-p.

3.2.1.9.3 High gain/A-band. The shaft of wing 2 or 4 shall move 1.64 ±0.17 deg p-p.

3.2.1.9.4 Low gain/A-band. The shaft of wing 2 or 4 shall move 0.72 ±0.08 deg p-p.

3.2.1.9.5 B-band. The shaft of wing 2 or 4 shall move 1.45 ±0.15 deg p-p.

3.2.1.9.6 C-band. The shaft of wing 2 or 4 shall move to 3.63 ±0.37 deg p-p.

3.2.1.9.7 High speed/D-band/prelaunch. The output at 16J1-9 shall be -1.8 ±1.0 Vdc, at J23-28 shall be +5 ±1 Vdc, at J23-9 shall be +5 ±1 Vdc and at J23-30 shall be -4.5 ±0.4 Vdc.

3.2.1.9.8 Missile velocity (Vm) high gain/D-band. The shaft of wing 2 or 4 shall move 2.17 ±0.22 deg p-p and the output at J21-26 shall be 31 ±1 Vp-p.

3.2.1.9.8.1 Vm medium gain/D-band. The following shall apply:

a. The output at J23-30 shall be -3.20 ±0.25 Vdc.

b. The shaft of wing 2 or 4 shall move 3.41 ±0.35 deg p-p.

3.2.1.9.8.2 Vm low gain/D-band. The following shall apply:

a. The output at J23-30 shall be -2.25 ±0.20 Vdc.

b. The shaft of wing 2 or 4 shall move 4.92 ±0.50 deg p-p.

3.2.1.9.9 Surface launch/3 Hz - 2 deg p-p roll yozzle. The shaft of wing 2 or 4 shall move 0.48 ±0.06 deg p-p and the output at J23-32 shall be +5 ±1 Vdc, and at J21-1 shall be -23.5 ±5.0 Vdc.

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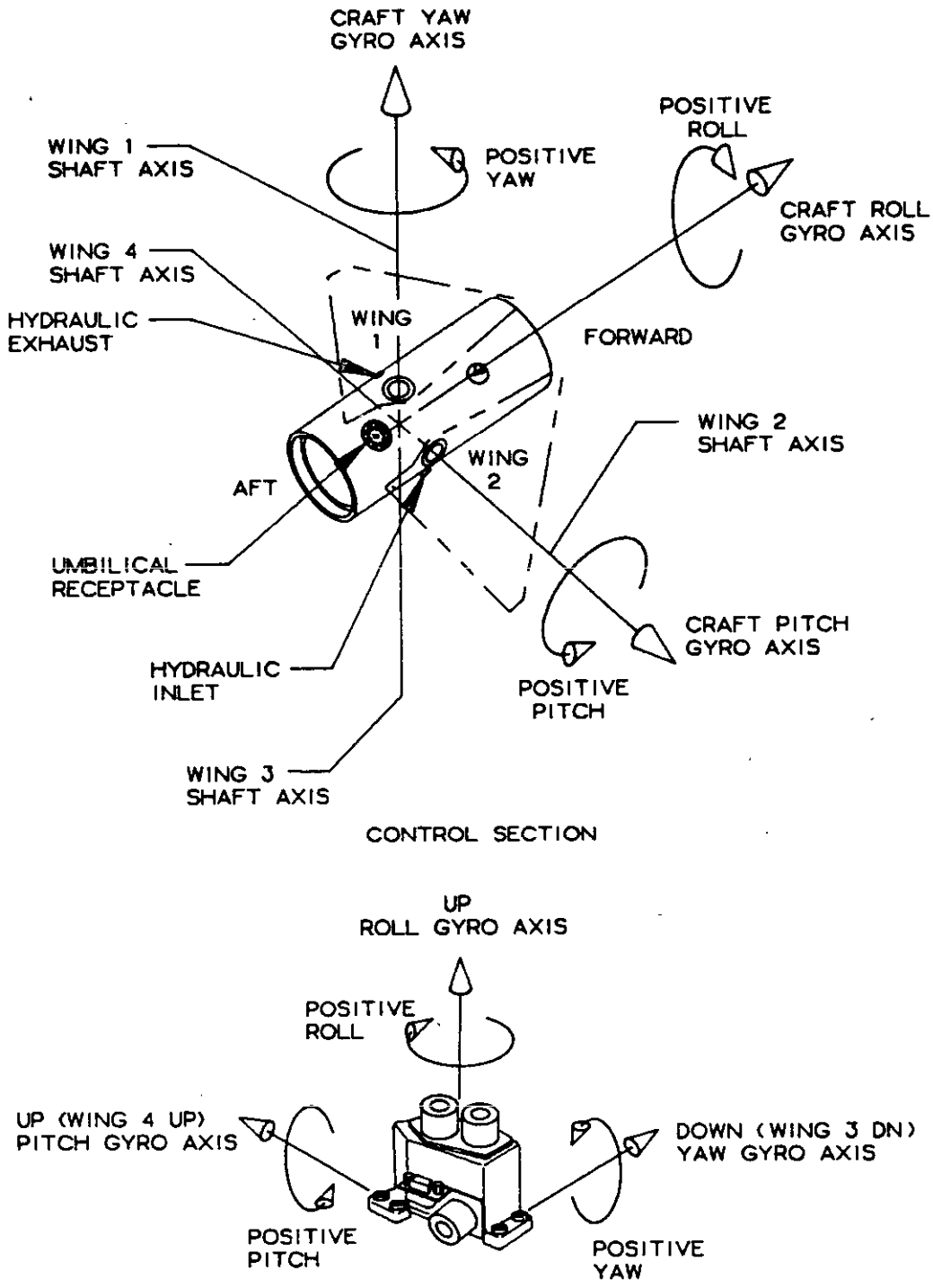


FIGURE 1. Craft gyro input axis orientation.

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3.2.1.9.10 Longitudinal accelerometer. The output at 16J1-9 shall change at a rate of -0.98 ±0.10 volts direct current/second (Vdc/s).

3.2.1.9.11 Roll command and timing. The output at 16J1-2 shall change -0.19 ±0.10 Vdc 0.60 ±0.06 s after launch indication is applied, and shall be -2.63 ±0.80 Vdc after 1.6 s, and shall stop changing (change not greater than 0.4 Vdc/s) in 3.0 ±0.3 s. The output at J23-34 shall change from +2.5 ±0.5 Vdc to +5 ±1 Vdc and the output at 16J1-11 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc, 0.6 ±0.1 s after J23-31 is changed to +5 Vdc. The output at 16J1-27 shall change from +2.5 ±0.5 Vdc to +5 ±1 Vdc and the output at 16J1-12 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc at 3.0 ±0.3 s after J23-31 is changed to +5 Vdc.

3.2.1.10 Yaw gyro gain/3 Hz - 2 deg p-p yaw yozzle.

3.2.1.10.1 Residual. The output at 16J1-6 shall be 0 ±50 mVdc.

3.2.1.10.2 B-band. The shaft of wing 1 (or 3) shall move 8.79 ±0.88 deg p-p, the outputs at 16J1-4 and 16J1-5 shall be not greater than 59 mVp-p and 75.5 mVp-p, respectively, and at 16J1-6 the 3 Hz output shall be 1.51 ±0.15 Vp-p and shall be positive for positive rotation (see Figure 1).

3.2.1.10.3 C-band. The shaft of wing 1 (or 3) shall move 15.73 ±1.58 deg p-p.

3.2.1.10.4 D-band. The shaft of wing 1 (or 3) shall move 21.5 ±2.2 deg p-p.

3.2.1.10.5 A-band. The shaft of wing 1 (or 3) shall move 3.96 ±0.40 deg p-p.

3.2.1.10.6 Surface launch. The shaft of wing 1 (or 3) shall move 3.22 ±0.32 deg p-p.

3.2.1.11 Yaw synthetic stability.

3.2.1.11.1 A-band/boost. The shaft of wing 1 (or 3) shall move 1.61 ±0.17 deg p-p.

3.2.1.11.2 A-band/glide. The shaft of wing 1 (or 3) shall move 0.79 ±0.09 deg p-p.

3.2.1.12 Pitch gyro gain/3 Hz - 2 deg p-p pitch yozzle.

3.2.1.12.1 Residual. The output at 16J1-5 shall be 0 ±50 mVdc.

3.2.1.12.2 B-band. The shaft of wing 2 or 4 shall move 8.79 ±0.88 deg p-p, the outputs at 16J1-6 and 16J1-4 shall be not greater than 75.5 mVp-p and 59 mVp-p, respectively, and at 16J1-5 the 3 Hz output shall be 1.50 ±0.15 Vp-p and shall be negative for positive rotation (see Figure 1).

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3.2.1.12.3 C-band. The shaft of wing 2 or 4 shall move 15.73 ±1.58 deg p-p.

3.2.1.12.4 D-band. The shaft of wing 2 or 4 shall move 21.5 ±2.2 deg p-p.

3.2.1.12.5 A-band. The shaft of wing 2 or 4 shall move 3.96 ±0.40 deg p-p.

3.2.1.12.6 Surface launch. The shaft of wing 2 or 4 shall move 3.22 ±0.32 deg p-p.

3.2.1.13 Pitch synthetic stability.

3.2.1.13.1 B-band/boost. The shaft of wing 2 or 4 shall move 3.01 ±0.31 deg p-p.

3.2.1.13.2 B-band/glide. The shaft of wing 2 or 4 shall move 1.48 ±0.16 deg p-p.

3.2.1.14 Guidance command loop (1 Hz 1.38 Vp-p input).

3.2.1.14.1 A-band/glide. The shaft of wing 1 (or 3) shall move 4.93 ±0.50 deg p-p. The shaft of wing 2 or 4 shall move 4.93 ±0.50 deg p-p.

3.2.1.14.2 B-band/boost. The shaft of wing 1 (or 3) shall move 9.20 ±0.92 deg p-p. The shaft of wing 2 or 4 shall move 9.20 ±0.92 deg p-p.

3.2.1.14.3 C-band. The shaft of wing 1 (or 3) shall move 13.9 ±1.4 deg p-p. The shaft of wing 2 or 4 shall move 13.9 ±1.4 deg p-p.

3.2.1.14.4 S-band. The shaft of wing 1 (or 3) shall move 5.78 ±0.58 deg p-p. The shaft of wing 2 or 4 shall move 5.78 ±0.58 deg p-p.

3.2.1.14.5 Pre-electronic time delay (ETD) inhibit. The shaft of wing 1 (or 3) shall move not greater than 0.15 deg p-p. The shaft of wing 2 or 4 shall move not greater than 0.15 deg p-p.

3.2.1.14.6 Alt Bit 2, TTG timing. The output at J23-9 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc not more than 50 microseconds (μs) after the input to J23-27 is changed to +5.0 ±0.5 Vdc.

3.2.1.14.7 Alt Bit 1, TTG timing. The output at J23-28 shall change from +5 ±1 Vdc to 0.0 ±0.5 Vdc not more than 50 μs after the input to J23-27 is changed to a ground.

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3.2.1.15 English bias (5 Vrms 10 Hz input).

3.2.1.15.1 Yaw. The 10 Hz output at J23-24 shall be 52.5 ±7.8 millivolts root-mean-square (mVrms).

3.2.1.15.2 Pitch. The 10 Hz output at J23-6 shall be 52.5 ±7.8 mVrms.

3.2.1.16 Autopilot outputs.

3.2.1.16.1 Wing lock/latch. The output at 16J1-29 shall be +12.8 ±1.3 Vdc.

3.2.1.16.2 Wing lock/latch (launch). The output at 16J1-29 shall be +4.16 ±0.40 Vdc.

3.2.1.17 Autopilot built-in-test (BIT).

3.2.1.17.1 Total BIT operation/tune mode. The output at J23-10 shall be 0.0 ±0.5 Vdc.

3.2.1.17.2 1500 Hz. The output at J23-10 shall be +5 ±1 Vdc.

3.2.1.17.3 Total BIT operation/prelaunch mode. The output at J23-10 shall be 0.0 ±0.5 Vdc for not greater than 2.5 s after full power is applied and then the output at J23-10 shall change to +5 ±1 Vdc.

3.2.1.17.4 1200 Hz. The output at J23-10 shall be 0.0 ±0.5 Vdc.

3.2.1.17.5 Head gyro sense. The output at J23-10 shall be +5 ±1 Vdc.

3.2.1.17.6 Wings.

3.2.1.17.6.1 Wings 1 and 3-right. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wings 1 and 3 are moved 2.00 ±0.25 deg off center.

3.2.1.17.6.2 Wings 1 and 3-left. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wings 1 and 3 are moved 2.00 ±0.25 deg off center.

3.2.1.17.6.3 Wing 2 up. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wing 2 is moved 2.00 ±0.25 deg off center.

3.2.1.17.6.4 Wing 2 down. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wing 2 is moved 2.00 ±0.25 deg off center.

3.2.1.17.6.5 Wing 4 up. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wing 4 is moved 2.00 ±0.25 deg off center.

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3.2.1.17.6.6 Wing 4 down. The output at J23-10 shall change from 0.0 ±0.5 Vdc to +5 ±1 Vdc when wing 4 is moved 2.00 ±0.25 deg off center.

3.2.1.18 Autopilot drifts.

3.2.1.18.1 Pitch and yaw drift/B-band. The outputs at 16J1-19 (pitch integrator test) and 16J1-20 (yaw integrator test) shall not change at a rate in excess of ±35 mVdc/s.

NOTE: 16R1 (pitch) and 16R2 (yaw) may be adjusted to satisfy this requirement.

3.2.1.18.2 Final roll drift/C-band. The average differential drift rate between wings 2 and 4 shall not exceed 0.1 degrees per second (deg/s).

NOTE: 16A3R13 may be adjusted to satisfy this requirement.

3.2.1.19 Accelerometer loops.

3.2.1.19.1 Yaw/1 gravity unit/glide mode-wing 4 up.

a. The output at 16J1-7 shall be +160 ±25 mVdc, at 16J1-8 shall be 0 ±25 mVdc, and at J23-25 shall be -164 ±25 mVdc.

b. The shaft wing 1 (or 3) shall move 8.97 ±0.89 deg/s. The final wing 1 (or 3) shaft position shall be 17.5 ±1.8 deg.

3.2.1.19.2 Yaw/1 gravity unit/glide mode-wing 2 up. The output at 16J1-7 shall be -160 ±25 mVdc. The average of the magnitude of the output at 16J1-7 with wing 4 up and then with wing 2 up shall be 160 ±16 mVdc.

3.2.1.19.3 Pitch/1 gravity unit/glide mode-wing 3 up.

a. The output at 16J1-8 shall be +160 ±25 mVdc, at 16J1-7 shall be 0 ±25 mVdc and at J23-26 shall be -164 ±25 mVdc.

b. The shaft of wing 2 or 4 shall move 8.97 ±0.89 deg/s. The final wing 2 or wing 4 shaft position shall be 17.5 ±1.8 deg.

3.2.1.19.4 Pitch/1 gravity unit/glide mode-wing 1 up. The output at 16J1-8 shall be -160 ±25 mVdc. The average of the magnitude of the output at 16J1-8 with wing 3 up and then with wing 1 up shall be 160 ±16 mVdc.

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3.2.1.20 Wing centering. The mechanical position of wings 1, 2, and 4 shall be 0.0 ±0.4 deg from zero reference.

NOTE: Zero reference of each wing is defined as a plane through the assembly roll axis and the wing shaft axis (see Figure 1). This plane is identified by a scribe mark located on the forward end of the control section housing (Drawing 479512).

3.2.1.21 Mode commands/surface to surface. The output at J21-3 shall be -1.0 ±0.2 Vdc and 1.0 ±0.2 Vrms.

3.2.1.21.1 Doppler acquisition mode select. The output at J21-4 shall be 240 ±50 mVrms.

3.2.1.22 Resistance measurements.

3.2.1.22.1 Squib resistance. The resistance between J20-J and J20-A and between J21-29 and J21-10 shall be 11.9 ±1.6 ohms.

3.2.1.22.2 Triggering device. The resistance between J21-2 and J21-9 shall be greater than 1 megohm (Mohm).

3.2.1.22.3 Video inject shield. The resistance between J21-5 and J21-9 shall be greater than 1 Mohm.

3.2.1.22.4 Continuity. The resistance between the indicated connections shall be not greater than 1 ohm:

a. J20-E and J21-23

b. J20-N and J21-21

c. J20-Z and J21-8

d. J20-k and J21-24

e. J20-h and J21-25

f. J20-b and J21-6

g. J20-F and J21-7

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h. J20-U and J21-22

i. J20-B and J20-a

j. J23-20 and J23-7

3.3 Design and construction.

3.3.1 Production drawings. The assembly shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on Drawing 917AS6200, 917AS9074 or 917AS9868, as applicable.

3.3.2 Standards of manufacture. As a minimum, the assembly shall be manufactured, inspected, and tested under conditions and procedures meeting the requirements specified herein.

3.3.2.1 Special working environment. The contractor shall provide adequate facilities for the fabrication, assembly, and inspection of supplies to be delivered in accordance with this specification. Unless otherwise specified in the contract or purchase order (see 6.2.1), the minimum standards for working environments of MIL-STD-1695 shall apply to the facility where final assembly, test, and delivery of the flight control assembly is accomplished.

3.3.2.2 Soldering. Preparation and procedures for soldering electrical connections shall be in accordance with WS-6536.

3.3.3 Workmanship. The assembly, including all parts and accessories, shall be constructed and finished in a manner that will assure compliance with all requirements of this specification. Except where there is a conflict with the applicable drawings, or unless otherwise specified herein, fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.1.1 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspections (see 4.2).

4.1.2 Test procedures, provisions, and equipment.

4.1.2.1 Calibration and maintenance. Unless otherwise specified herein, all test equipment shall be calibrated and maintained in accordance with the provisions of MIL-STD-45662.

4.1.2.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall be accurate to within one-fifth of the tolerance given for the variable to be measured, except that the phase measuring apparatus shall be accurate to within 1 deg. In cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within five percent of that requirement except that the open-ended resistance requirements of 3.2.1.22.4 shall be accurate within 0.5 ohm. The rate measurements of 3.2.1.9.10, 3.2.1.9.11 and 3.2.1.19 may be made using test equipment accurate to within ±3 percent.

4.1.2.3 Test equipment loading. All test equipment used for the measurements specified in 4.2.2 shall meet the requirements of Table II. Mechanical load on wing shafts shall be less than 50 pound-feet.

4.1.2.4 Test facility environment. Unless otherwise specified herein, tests shall be performed under the following conditions of environment.

a.	Temperature:	Ambient, between +18.3 and +35 degrees Celsius (°C).
b.	Altitude:	Elevation of test facility.
c.	Relative humidity	Maintained between 30 and 70 percent.

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TABLE II. Test equipment loading.

Test point designation	Minimum test equipment resistive loading	Maximum value of test equipment capacitance
J20-D	100 kohms	2000 pF
J20-S	100 kohms	1/
J20-T	500 kohms	1/
J20-V	500 kohms	1/
J21-3	10 kohms	0.1 µF
J21-4	10 kohms	0.1 µF
J21-13	100 kohms	1/
J21-15	2/	1/
J21-17	2/	1/
J21-18	1 kohm	0.1 µF
J21-19	1 kohm	0.1 µF
J21-26	1 kohm	1 µF
J21-27	100 kohms	1/
J21-32	100 kohms	1/
J21-33	2/	1/
J21-35	2/	1/
J21-37	1 kohm	0.1 µF
J22-15	100 kohms	5000 pF
J22-34	100 kohms	5000 pF
J23-1	10 kohms	10 µF
J23-2	10 kohms	10 µF
J23-3	10 kohms	10 µF
J23-5	2 kohms	0.1 µF
J23-6	2 kohms	0.1 µF
J23-9	20 kohms	1/
J23-10	10 kohms	1/
J23-11	10 kohms	1/
J23-14	10 kohms	1/
J23-15	10 kohms	1/
J23-16	10 kohms	1/
J23-24	2 kohms	0.1 µF
J23-25	2 kohms	0.1 µF
J23-26	2 kohms	0.1 µF
J23-28	20 kohms	1/
J23-30	2 kohms	0.1 µF
J23-32	20 kohms	1/
J23-34	20 kohms	1/
16J1-1	1 Mohm	1/
16J1-2	1 Mohm	1/
16J1-3	1 Mohm	1/
16J1-4	2 Mohms	1/

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TABLE II. Test equipment loading. - Continued.

Test point designation	Minimum test equipment resistive loading	Maximum value of test equipment capacitance
16J1-5	1 Mohm	1/
16J1-6	1 Mohm	1/
16J1-7	1 Mohm	1/
16J1-8	1 Mohm	1/
16J1-9	1 Mohm	1/
16J1-10	1 Mohm	1/
16J1-11	1 Mohm	0.1 µF
16J1-12	1 Mohm	1/
16J1-19	1 Mohm	0.5 µF
16J1-20	1 Mohm	1/
16J1-21	1 Mohm	0.5 µF
16J1-24	100 kohms	1/
16J1-25	100 kohms	1/
16J1-27	1 Mohm	1/
16J1-29	1 Mohm	1/

1/ Indicates that only dc measurements are made at these points. The value of test equipment capacitance for these points will limit the speed of measurement only and not the measurement accuracy.

2/ Indicates that the test point has a test load applied as specified in 4.1.4.8. Test equipment resistance and capacitance shall be of proper value to maintain the loading conditions of 4.1.4.8.

4.1.2.5 Common ground. Unless otherwise specified herein, the assembly under test and all test equipment shall be referenced to a common ground. Common ground shall be J23-7, J21-9 and J20-A.

4.1.2.6 Input tolerance. Unless otherwise specified herein, all inputs shall be maintained within ±1 percent of the specified value.

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4.1.3 General instructions. Unless otherwise specified herein, the general instructions specified in 4.1.3.1 through 4.1.3.4 shall apply to the performance characteristics tests of 4.2.2.

4.1.3.1 External hydraulic power. When specified in 4.2.2, the assembly shall be provided with hydraulic fluid in accordance with MIL-H-5606. The hydraulic fluid shall be supplied from an external source to the high pressure inlet port indicated in Figure 1. The external source shall be capable of supplying and maintaining the hydraulic fluid to the inlet port at 3,200 ±100 pounds per square inch gauge.

4.1.3.1.1 Hydraulic fluid temperature. Unless otherwise specified herein, the external hydraulic fluid supply shall be maintained at a temperature between +13°C and +41°C.

4.1.3.1.2 Hydraulic fluid application. The hydraulic fluid pressure shall not be applied to the assembly prior to the application of electrical power (see 4.1.4.1).

4.1.3.1.3 Hydraulic fluid contaminant level. The hydraulic fluid solid particle contamination at the inlet port shall be not greater than the limits specified in Table III.

TABLE III. Maximum acceptance level of solid particle contamination (number of solid particles per 100 milliliter (ml) sample).

Particle size (microns)	Count per 100 ml
5 - 10	5,000
10 - 25	1,500
25 - 50	250
50 - 100	30
100 - 300	3
over - 300	0

4.1.3.1.4 Wing shaft receptacles. The hydraulic fluid pressure shall not be applied to the assembly prior to the insertion of a simulated wing stud (see Drawing 917AS213, 917AS216, or equal) into each of the wing stud receptacles.

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4.1.3.2 Measurement of wing shaft motion. Test requirements to measure wing shaft position and wing shaft motion may be satisfied by either measuring the wing shaft position and motion directly or by monitoring the output of the wing follow-up potentiometers at 16J1-1 (wings 1 and 3), 16J1-2 (wing 2), and 16J1-3 (wing 4) using a scale factor of 0.4 Vdc per deg. A positive voltage indicates a deflection of wings 1 and 3 to the right and a deflection of wing 2 or wing 4 up. All sinusoidal measurements shall refer to the fundamental frequency of the input drive signal. The alternative of measuring wing shaft position by means of the follow-up potentiometers does not apply to the requirements for wing centering (see 3.2.1.20). In this case, the wing shaft position shall be determined directly. The alternative of measuring wing shaft motion directly shall not apply to those requirements specifying wing shaft motion less than 4 deg p-p. During the tests of 4.2.2 requiring the measurement of sinusoidal wing motions, the pitch and yaw integrators may be reset prior to the measurement by applying a momentary ground to J20-H.

4.1.3.3 Control section orientation. The assembly shall be placed on a test stand suitable to support the assembly in a convenient orientation. When specified in 4.2.2, the assembly or gyroscope assembly (unit 14) (see Figure 1) shall be rotated about the craft yaw gyro axis (yaw yozzle), pitch gyro axis (pitch yozzle), or roll gyro axis (roll yozzle) with a 2.00 ±0.04 deg p-p sinusoidal motion at a frequency of 3 Hz. The assembly shall also be oriented to umbilical up, wing 3 up and wing 4 up for the accelerometer test.

4.1.3.4 Test sequence. Unless otherwise specified herein, each test paragraph is structured to be independent of the residual test conditions from the preceding test and refers only to exception to the test conditions of 4.1.4.5. Consequently, some latitude may be exercised in selecting the most expedient test sequence, provided that the drift minimization tests (see 4.2.2.4, 4.2.2.5 and 4.2.2.18) are conducted in the order shown.

4.1.4 Test conditions. Unless otherwise specified herein, the test condition specified in 4.1.4.1 through 4.1.4.9 shall apply to the performance characteristics tests of 4.2.2.

4.1.4.1 DC/pick off supply voltage. When specified in 4.2.2, the assembly shall be supplied with the following inputs to the contacts specified:

	Contact No.	Vdc	Approximate milliamperes (mA)
a.	J21-32 1/	+26.5 ±0.5	2800

Applied through a 1N250A diode, or equivalent (cathode towards J21-32). The ac ripple shall be not greater than 100 millivolts true root-mean-square (mVtrms).

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b. J21-13 1/ -26.5 ±0.5 2800

Applied through a 1N250A diode, or equivalent (anode towards J21-13). The ac ripple shall be not greater than 100 mVtrms.

c. J23-14 2/ +20.6 ±0.5 210

Applied through a 1N250A diode, or equivalent (cathode towards J23-14). The ac ripple shall be not greater than 100 mVtrms.

d. J23-11 2/ -20.6 ±0.5 240

Applied through a 1N250A diode, or equivalent (anode towards J23-11). The ac ripple shall be not greater than 100 mVtrms.

e. 9.0 Vrms at 1500 Hz, 750 mA peak current applied between J21-30 (high) and J21-11 (low).

1/ The input shall be returned via J21-14.

2/ The input shall be returned via J23-12.

4.1.4.2 Control section 400 Hz power. When specified in 4.2.2, 400 Hz, three phase, wye connected power shall be specified as follows:

a. 113.0 ±6.5 Vrms, 400 ±20 Hz, phase 1 applied to J20-j.

b. 113.0 ±6.5 Vrms, 400 ±20 Hz, phase 2 applied to J20-W.

c. 113.0 ±6.5 Vrms, 400 ±20 Hz, phase 3 applied to J20-X.

d. Neutral connected to common ground J20-B.

4.1.4.3 Input signals. When specified herein, the assembly shall be supplied with the following inputs to the contacts specified:

a. 5.84 Vrms at 1200 Hz phase A between J21-31 and J21-14.

b. 5.84 Vrms at 1200 Hz phase B between J21-12 and J21-14. This voltage shall be 90 ±5 deg phase shifted (lag) with respect to a above.

c. 0 to 35 Vp-p at 400 ±20 Hz applied to J20-Y as specified in 4.2.2.

d. 0.0 to 0.6 Vrms at 1200 Hz applied between J23-36 and J23-37 as specified in 4.2.2.

4.1.4.4 Connections. The connections specified in Table IV are required to conduct the specified tests.

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TABLE IV. Connections.

Function	From	To
System ground to A12 power return	J22-9	J21-34
+26.5 Vdc	J23-19	J21-32
-26.5 Vdc	J23-17	J21-13
±26.5 return	J23-18	J21-14
+43.5 Vdc	J32-23	J22-17
+43.5 return	J32-24	J22-16
Pitch accel pick off voltage	J32-28	J22-29
Yaw accel pick off voltage	J32-9	J22-11
Accel P.O. return	J32-27	J22-30
System ground	J32-30	J22-9
400 Hz return	J32-22	J22-36
400 Hz phase 1	J32-1	J22-19
400 Hz phase 2	J32-20	J22-37
400 Hz phase 3	J32-21	J22-18
Craft inst (1500 Hz) Hi	J32-26	J22-28
Craft inst (1500 Hz) Lo	J32-10	J22-10
Craft gyro current sense Hi	J32-4	J22-12
Craft gyro current sense Lo	J32-29	J22-31
Pitch gyro P.O.	J32-33	J22-24
Yaw gyro P.O.	J32-14	J22-6
Roll gyro P.O.	J32-32	J22-25
Craft gyro P.O. return	J32-13	J22-7
Latch command	J32-19	J22-1
PD/CW select	J32-35	J22-22
Pitch head aim/TAS	J32-31	J22-26
Yaw English bias	J32-37	J22-20
Pitch English bias	J32-18	J22-2
Alt No. 1	J32-36	J22-21
Alt No. 2	J32-17	J22-3
Surface launch command	J32-16	J22-4
Wing lock test	J32-34	J22-23
Alt No. 3	J32-6	J22-14
Roll command	J32-8	J22-33
Missile ready, front and rear	J32-25	J22-32
BIT signal, head gyro (lo)	J32-7	J23-37

4.1.4.5 Initial test conditions. The initial test conditions shall be as specified in Table V.

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TABLE V. Initial test conditions.

Contact	Condition	Function
16J1-17	Open	Pitch gyro amp test
16J1-18	Open	Yaw gyro amp test
16J1-22	Ground	Yaw test point
16J1-23	Ground	Pitch test point
16J1-26	Ground	Roll test point
16J1-27	Ground	Boost test
16J1-28	Ground	Launch +0.6 s test
16J1-30	Open	Enable latch test
16J1-34	Open	Tune test point
16J1-35	Open	Craft Bit test
J20-C	Open	Surface launch command
J20-D (Type I)	Open	Alt No. 3
J20-D (Type II)	18 ±1 kohm resistance (100 mW minimum)	Alt No. 3
J20-H	Ground	Wing lock test
J20-K	Ground	Roll command
J20-L	Open	Yaw gyro test
J20-M	Open	Pitch gyro test
J20-R	Ground	Yaw English bias
J20-U	Open	PD/CW command
J20-c	Ground	Pitch English bias
J20-d	Open	Alt No. 1
J20-e	Open	Alt No. 2
J21-28	-6.0 ±0.5 Vdc through a resistance of 7 ±1 kohms (20 milliwatt (mW) minimum)	Gain switch
J23-4	Ground	A/P command return
J23-8	Ground	Long accel demod
J23-10	+5 ±1 Vdc through a 2.0 ±0.2 kohms resistance (50 mW minimum)	A/P BIT
J23-13	Open	SSLO temp sense
J23-21	Open	Enable command
J23-22	Short to J23-4	A/P pitch command
J23-23	Short to J23-4	A/P yaw command
J23-27	+5.0 ±0.5 Vdc	TTGO (post TTG logic)
J23-29	Ground	K9 switch
J23-31	Ground	Launch switch
J23-33	Ground	Ready indicate
J23-35	Open	Battery status

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4.1.4.6 Control section full power. When specified in 4.2.2, the following sequence shall be followed to apply full power (observe CAUTION which follow):

a. Apply the dc/pick off supply power as specified in 4.1.4.1.

b. With an open on J20-U, apply the control section 400 Hz power as specified in 4.1.4.2.

c. Apply -6.0 ±0.5 Vdc through a 7 ±1 kilohm (kohm) (20 mW minimum), resistance to J21-28 as specified in 4.1.4.5.

d. Apply the input signal as specified in 4.1.4.3a and b.

e. Remove the 400 Hz power 2.0 ±0.2 s after applying the inputs as specified in 4.1.4.3a and b. Not more than 10 ms after the 400 Hz power is removed simultaneously change the input to J21-28 to +14.7 ±1.2 Vdc, through a 7 ±1 kohm (20 mW minimum), resistance.

CAUTION

1. A time interval of at least 15 s shall be allowed after the removal of the input signals of d above repeating the above sequence to apply full power.

2. The above sequence to apply full power shall not be repeated if full power is already applied.

3. During the tests of 4.2.2, remove all power from the control section if the dc output between J22-15 and J22-16 or between J22-34 and J22-16 exceeds ±1.5 Vdc.

4.1.4.7 Surface launch mode selection. When specified in 4.2.2, surface launch mode shall be selected by applying -23.5 ±5.0 Vdc to J20-C.

4.1.4.8 Loads.

a. Power supply. The loads specified in Table Vl shall be applied during the tests of 4.2.2.1 and 4.2.2.3 only except the loads of J21-17 and J21-35 shall be applied during the tests of 4.2.2.2.1, only. The loads shall be applied between the indicated contacts and ground.

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TABLE VI. Power supply loads.

Contact	Function	Loads
J21-17	+43.3 Vdc	17.4 ±2.0 ohms (150 W minimum)
J21-27	+53 Vdc	22.0 ±2.5 ohms (150 W minimum)
J21-35	-43.5 Vdc	17.4 ±2.0 ohms (150 W minimum)
J23-11	-20.6 Vdc	4.2 ±0.5 ohms (150 W minimum)
J23-14	+20.6 Vdc	4.2 ±0.5 ohms (150 W minimum)
J23-17	-26.5 Vdc	88 ±8 ohms (10 W minimum)
J23-19	+26.5 Vdc	88 ±8 ohms (10 W minimum)

b. Autopilot outputs. Connect a 10.0 ±0.5 kohms (50 mW minimum) resistance between each of the indicated contacts and ground, as specified in Table VII.

c. Filtered surface-to-surface mode command. Connect a 1.00 ±0.05 kohms (10 mW minimum) resistance between contact J21-3 and ground.

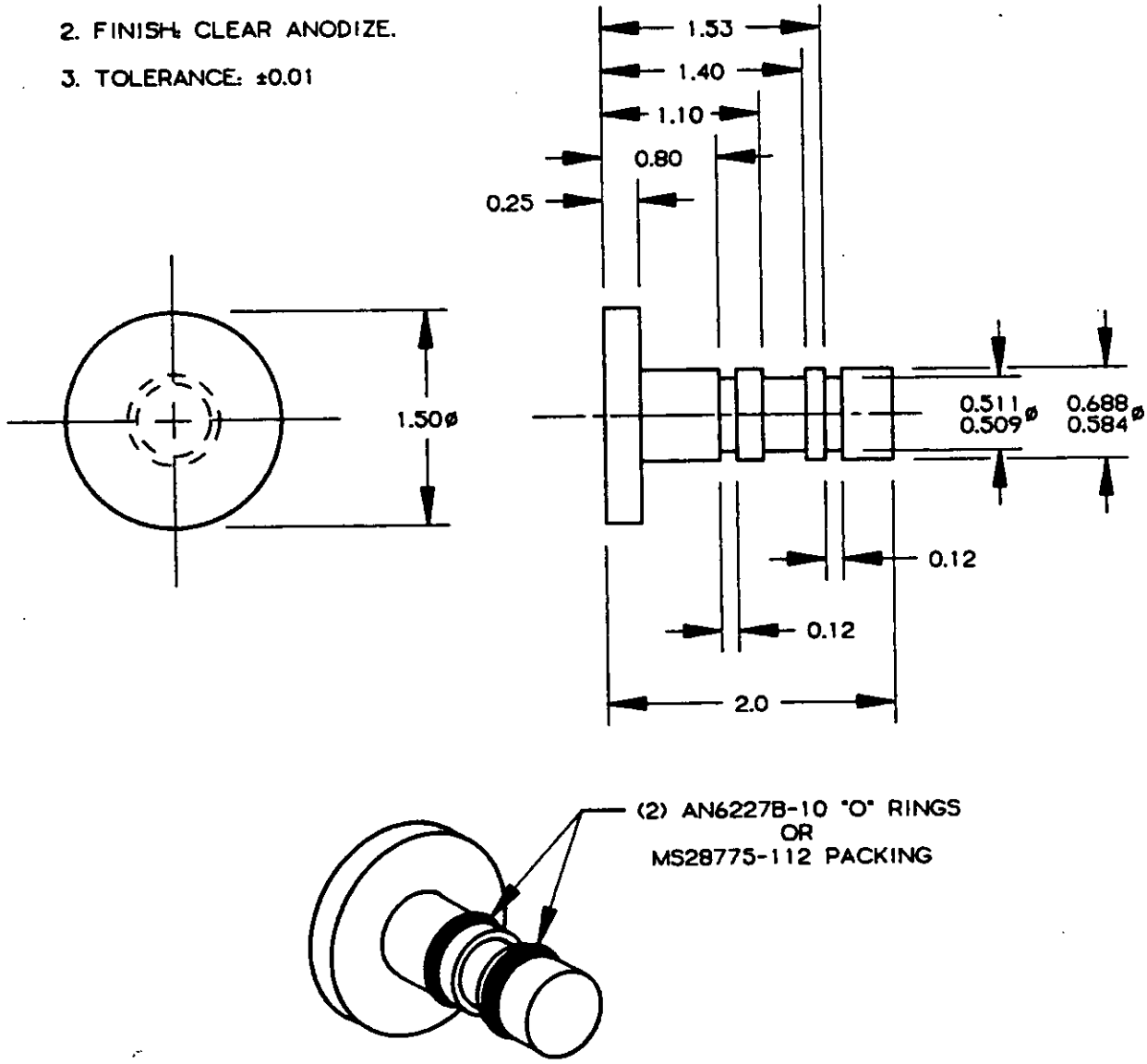
TABLE VII. Autopilot outputs.

Contact	Function
J20-f	Latch command
J23-5	Roll rate
J23-6	Pitch English bias (scaled)
J23-9	Alt Bit 2
J23-24	Yaw English bias (scaled)
J23-25	Yaw accel demod
J23-26	Pitch accel demod
J23-28	Alt Bit 1
J23-32	"S" command (surface launch command latch)

4.1.4.9 Accumulator and flow valve assembly. The accumulator and flow valve assembly (see Drawing 917AS8148) need not be connected except when conducting the test of 4.2.2.22. A hydraulic test plug (see Figure 2) shall be used in the absence of the accumulator and flow valve assembly during the conduct of the tests of 4.2.2.6 through 4.2.2.20.

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1. MATERIAL: 1.5 DIA 2024T4 AL ROD
2. FINISH: CLEAR ANODIZE.
3. TOLERANCE: ±0.01



NOTE:

THIS FIGURE IS PROVIDED FOR REFERENCE ONLY. ANY FUNCTIONALLY EQUIVALENT ITEM SUITED FOR THE PURPOSE OF SEALING THE HYDRAULIC RECEPTICLE VACATED BY THE REMOVAL OF THE ACCUMULATOR AND FLOW VALVE ASSEMBLY (30003-917AS8148) DURING CONTROL SECTION TESTING IS ACCEPTABLE.

FIGURE 2. Hydraulic test plug.

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4.2 Quality conformance inspections. Quality conformance inspections shall consist of visual and mechanical inspection and performance characteristics tests to verify conformance with the requirements stated herein.

4.2.1 Visual and mechanical examination. Each assembly shall be inspected for conformance to the requirements of 3.3.

4.2.2 Electrical performance tests. Each assembly shall be tested in accordance with the test conditions of 4.1.4 to verify conformance to the corresponding requirements of 3.2.1. Failure of any assembly to meet any of the test requirements shall be cause for rejection of the assembly.

4.2.2.1 AC/DC conversion. With 400 Hz power of 4.1.4.2 applied, +43.5 ±4.0 Vdc applied to J23-13, and ground applied to J20-U, the assembly shall meet the requirements of 3.2.1.1.

4.2.2.2 Gyro signals.

4.2.2.2.1 Over voltage condition. With the full power of 4.1.4.6 applied, the assembly shall meet the requirements of 3.2.1.2.1.

4.2.2.2.2 Normal run condition. With the full power of 4.1.4.6 applied, the assembly shall meet the requirements of 3.2.1.2.2 after the 400 Hz power is removed as specified in 4.1.4.6e.

4.2.2.3 Prelaunch tuning. The test conditions specified below shall carry over from a to b to c to d:

a. No signal. With the 400 Hz power of 4.1.4.2 applied, then +43.5 ±4.0 Vdc applied to J23-13, and ground applied to J20-U, the assembly shall meet the requirements of 3.2.1.3a.

b. No retune. With the ground removed from J20-U for not less than 5 s and then reapplied, the assembly shall meet the requirement of 3.2.1.3b.

c. No radar on. With the 400 Hz power of 4.1.4.2 removed for at least 5 s, the ground removed from J20-U, and then 400 Hz power of 4.1.4.2 reapplied, the assembly shall meet the requirements of 3.2.1.3c.

d. Enable command. With +26.5 ±1.0 Vdc applied for no more than 2 s to J23-21, the assembly shall meet the requirements of 3.2.1.3d.

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4.2.2.3.1 Prelaunch tune. With the 400 Hz power of 4.1.4.2 applied, +26.5 ±0.5 Vdc applied to J23-35 and the input to J23-33 changed from a ground to +4.5 ±0.1 Vdc, the assembly shall meet the requirements of 3.2.1.3.1.

CAUTION

The conditions of 4.2.2.3a, 4.2.2.3d and 4.2.2.3.1 shall be applied for not more than 5 s with an "on"/"off" duty cycle of not more than 10 percent.

4.2.2.3.2 Missile ready latch. Under the final conditions of 4.2.2.3.1 except with the input to J23-33 changed to a ground, and then not less than 10 s later +43.5 Vdc applied to J23-13, a ground applied to J20-U, and after a duration of not less than 2.5 s, the assembly shall meet the requirements of 3.2.1.3.2.

4.2.2.4 Roll drift. With the 400 Hz power of 4.1.4.2 applied for not less than 3 minutes, and then dc/pick off supply power of 4.1.4.1 applied and the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.4.

4.2.2.5 Pitch and yaw launch drift/C-band. With the 400 Hz power of 4.1.4.2 applied for not less than 3 minutes and then dc/pick off supply power of 4.1.4.1 applied, a ground applied to J20-d and J20-D, ground removed from J20-H, 16J1-22, 16J1-23, and 16J1-28, the input to J23-29 changed to +5 ±1 Vdc, and the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.5.

NOTES:

1. The 400 Hz power of 4.1.4.2 shall be applied for not less than 3 minutes for the heaters to warm-up before continuing this sequence and the application of 4.1.4.6.

2. Allow not less than 1 s but not more than 30 s after the input to J23-31 is changed to +5 Vdc before verifying conformance to 3.2.1.5.

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4.2.2.6 Wing servos 1 and 3. With the dc/pick off supply power of 4.1.4.1 applied and external hydraulic pressure applied as specified in 4.1.3.1, a 10 Hz input shall be applied through a 53.6 ±1.8 kohms (10 mW minimum) resistance to 16J1-14 at a level sufficient to obtain a wing 1 motion of 1.00 ±0.02 deg p-p, the assembly shall meet the requirement of 3.2.1.6.

4.2.2.6.1 Gain margin. Under the conditions of 4.2.2.6 with wings (see Drawing 917AS754 or equal) inserted in the wing 1 and 3 locations, the input to 16J1-14 removed and a 17.4 ±0.4 kohms (10 mW minimum) resistance connected between 16J1-1 and 16J1-14 and then a +4.0 ±0.4 V, 40 ±20 ms pulse (with a 10 ms maximum rise time) applied through a 20 ±1 kohms (10 mW minimum) resistance to 16J1-14, the assembly shall meet the requirement of 3.2.1.6.1.

4.2.2.7 Wing servo 2. With the dc/pick off supply power of 4.1.4.1 applied and external hydraulic pressure applied as specified in 4.1.3.1, a 10 Hz input shall be applied through a 53.6 ±1.8 kohms (10 mW minimum) resistance to 16J1-15 at a level sufficient to obtain a wing 2 motion of 1.00 ±0.02 deg p-p, the assembly shall meet the requirement of 3.2.1.7.

4.2.2.7.1 Gain margin. Under the conditions of 4.2.2.7, with a wing (see Drawing 917AS754 or equal) inserted in wing 2 location, the input to 16J1-15 removed and a 17.4 ±0.4 kohms (10 mW minimum) resistance connected between 16J1-2 and 16J1-15 and then a +4.0 ±0.4 V, 40 ±20 ms pulse (with a 10 ms maximum rise time) applied through a 20 ±1 kohm (10 mW minimum) resistance to 16J1-15, the assembly shall meet the requirement of 3.2.1.7.1.

4.2.2.8 Wing servo 4. With the dc/pick off supply power of 4.1.4.1 applied and external hydraulic pressure applied as specified in 4.1.3.1, a 10 Hz input shall be applied through a 53.6 ±1.8 kohms (10 mW minimum) resistance to 16J1-16 at a level sufficient to obtain a wing 4 motion of 1.00 ±0.02 deg p-p, the assembly shall meet the requirement of 3.2.1.8.

4.2.2.8.1 Gain margin. Under the conditions of 4.2.2.8 with a wing (see Drawing 917AS754 or equal) inserted in the wing 4 location, the input to 16J1-16 removed and 17.4 ±0.4 kohms (10 mW minimum), resistance connected between 16J1-3 and 16J1-16, and then a +4.0 V, 40 ±20 ms pulse (with a 10 ms maximum rise time) applied through a 20 ±1 kohms (10 mW minimum), resistance to 16J1-16, the assembly shall meet the requirement of 3.2.1.8.1.

4.2.2.9 Roll autopilot/3 Hz - 2 deg p-p roll yozzle.

4.2.2.9.1 Residual. With the full power of 4.1.4.6 applied, the assembly shall meet the requirement of 3.2.1.9.1.

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4.2.2.9.2 Demodulators. With the full power of 4.1.4.6 applied, external hydraulic pressure applied as specified in 4.1.3.1, ground removed from 16J1-26, a ground applied to 16J1-17 and 16J1-18, and a roll yozzle as specified in 4.1.3.3 applied, the assembly shall meet the requirement of 3.2.1.9.2.

4.2.2.9.3 High gain/A-band. Under the final conditions of 4.2.2.9.2, except with the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.3.

4.2.2.9.4 Low gain/A-band. Under the final conditions of 4.2.2.9.2, except with a ground applied to J23-27, ground removed from 16J1-28, and J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.4.

4.2.2.9.5 B-band. Under the final conditions of 4.2.2.9.2, except with ground removed from 16J1-28, the input to J23-31 changed to +5 ±1 Vdc and a ground applied to J20-e, the assembly shall meet the requirements of 3.2.1.9.5.

4.2.2.9.6 C-band. Under the final conditions of 4.2.2.9.2, except with a ground removed from 16J1-28, the input to J23-31 changed to +5 ±1 Vdc and a ground applied to J20-d and J20-D, the assembly shall meet the requirements of 3.2.1.9.6.

4.2.2.9.7 High speed/D-band/prelaunch. Under the final conditions of 4.2.2.9.2, except with the input to J20-Y changed to 31 ±1 Vp-p at 400 ±20 Hz and a ground applied to J20-e, J20-d and J20-D, the assembly shall meet the requirements of 3.2.1.9.7.

4.2.2.9.8 Vm high gain/D-band. Under the final conditions of 4.2.2.9.2, except with the input to J20-Y changed to 31 ±1 Vp-p at 400 ±20 Hz, ground removed from 16J1-28, ground applied to J20-e, J20-d and J20-D, +26.5 ±0.5 Vdc applied to 16J1-30, then open J20-e, J20-d, and J20-D, and then the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.8.

4.2.2.9.8.1 Vm medium gain/D-band. Under the final conditions of 4.2.2.9.8, except with the input to J23-31 changed to a ground and the input to J20-Y changed to 22.0 ±0.7 Vp-p at 400 ±20 Hz, the assembly shall meet the requirements of 3.2.1.9.8.1a. With the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.8.1b.

4.2.2.9.8.2 Vm low gain/D-band. Under the final conditions of 4.2.2.9.8.1, except with the input to J23-31 changed to a ground and the input to J20-Y changed to 15.5 ±0.5 Vp-p at 400 ±20 Hz, the assembly shall meet the requirements of 3.2.1.9.8.2a. With the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.8.2b.

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4.2.2.9.9 Surface launch. Under the final conditions of 4.2.2.9.2, except with surface launch mode selected as specified in 4.1.4.7 and the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.9.9.

4.2.2.9.10 Longitudinal accelerometer. Under the final conditions of 4.2.2.9.2, except with +2.3 ±0.1 Vdc applied to J23-8, the input to J20-Y changed to 31 ±1 Vp-p and the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirement of 3.2.1.9.10.

4.2.2.9.11 Roll command timing. Under the final conditions of 4.2.2.9.2, except with surface launch mode selected as specified in 4.1.4.7, no roll yozzle applied, the grounds removed from 16J1-27, J20-H and 16J1-28, the input to J20-K changed to +18.8 Vdc and then the input to J23-31 changed to +5 ±1 Vdc (launch indication), and the input to J20-K removed simultaneously, the assembly shall meet the requirements of 3.2.1.9.11.

4.2.2.10 Yaw gyro gain/3 Hz - 2 deg p-p yaw yozzle.

4.2.2.10.1 Residual. With the full power of 4.1.4.6 applied, the assembly shall meet the requirement of 3.2.1.10.1.

4.2.2.10.2 B-band. Under the condition of 4.2.2.10.1, external hydraulic pressure shall be applied as specified in 4.1.3.1. A ground shall be applied to J20-e, ground applied to J23-27 and a yaw yozzle as specified in 4.1.3.3 shall be applied, the assembly shall meet the requirements of 3.2.1.10.2.

4.2.2.10.3 C-band. Under the final conditions of 4.2.2.10.2, and with a ground applied to J20-D and J20-d, the input to J23-27 changed to ground, the assembly shall meet the requirement of 3.2.1.10.3.

4.2.2.10.4 D-band. Under the final conditions of 4.2.2.10.3 and with a ground applied to J20-e and the input to J23-27 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.10.4.

4.2.2.10.5 A-band. Under the final conditions of 4.2.2.10.2, with the ground removed from J20-e, the assembly shall meet the requirement of 3.2.1.10.5.

4.2.2.10.6 Surface launch. Under the final conditions of 4.2.2.10.2, with the ground removed from J20-e, the surface launch mode selected as specified in 4.1.4.7, the assembly shall meet the requirement of 3.2.1.10.6.

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4.2.2.11 Yaw synthetic stability.

4.2.2.11.1 A-band/boost. With the full power of 4.1.4.6 applied and external hydraulic pressure applied as specified in 4.1.3.1 and a yaw yozzle applied as specified in 4.1.3.3, a ground applied to 16J1-18, with the ground removed from J20-H and 16J1-22 and +5 ±1 Vdc applied to J23-31, the assembly shall meet the requirement of 3.2.1.11.1.

4.2.2.11.2 A-band/glide. Under the final conditions of 4.2.2.11.1, with the ground removed from J23-29 and +5 ±1 Vdc applied to J23-29, the assembly shall meet the requirement of 3.2.1.11.2.

NOTE: It may be necessary, during the performance of 4.2.2.11, to further adjust resistor 16R2 to minimize yaw drift and assure conformance to 3.2.1.5.

4.2.2.12 Pitch gyro gain/3 Hz - 2 deg p-p pitch yozzle.

4.2.2.12.1 Residual. With the full power of 4.1.4.6 applied, the assembly shall meet the requirement of 3.2.1.12.1.

4.2.2.12.2 B-band. Under the final conditions of 4.2.2.12.1, external hydraulic pressure shall be applied as specified in 4.1.4.1 and a pitch yozzle shall be applied as specified in 4.1.4.3. With a ground applied to J20-e, the assembly shall meet the requirement of 3.2.1.12.2.

4.2.2.12.3 C-band. Under the final conditions of 4.2.2.12.2, with a ground applied to J20-d and J20-D, and the input to J23-27 changed to a ground, the assembly shall meet the requirements of 3.2.1.12.3.

4.2.2.12.4 D-band. Under the final conditions of 4.2.2.12.3 and with a ground applied to J20-e, and the input to J23-27 changed to +5.0 ±1.0 Vdc, the assembly shall meet the requirements of 3.2.1.12.4.

4.2.2.12.5 A-band. Under the final conditions of 4.2.2.12.4 with the grounds removed from J20-D, J20-e and J20-d, the assembly shall meet the requirements of 3.2.2.12.5.

4.2.2.12.6 Surface launch. Under the final conditions of 4.2.2.12.2, the ground removed from J20-e, surface launch mode selected as specified in 4.1.4.7, the assembly shall meet the requirements of 3.2.1.12.6.

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4.2.2.13 Pitch synthetic stability.

4.2.2.13.1 B-band/boost. With the full power of 4.1.4.6 applied and with external hydraulic pressure applied as specified in 4.1.3.1, ground applied to J20-D and 16J1-17, ground removed from J20-H and 16J1-23, a pitch yozzle shall be applied as specified in 4.1.3.3. With +5 ±1 Vdc applied to J23-31, the assembly shall meet the requirements of 3.2.1.13.1.

4.2.2.13.2 B-band/glide. Under the conditions of 4.2.2.13.1, with the ground removed from J23-29 and +5 ±1 Vdc applied to J23-29, the assembly shall meet the requirements of 3.2.1.13.2.

NOTE: It may be necessary, during the performance of 4.2.2.13, to further adjust resistor 16R1 to minimize pitch drift and assure conformance to 3.2.1.5.

4.2.2.14 Guidance command loop (1 Hz 1.38 Vp-p input).

4.2.2.14.1 A-band/glide. With the dc power of 4.1.4.1 applied, external hydraulic pressure shall be applied as specified in 4.1.3.1. With the inputs to J20-e and J20-D changed to a ground, the inputs to J23-23 and J23-22 changed to 1.38 Vp-p at 1 Hz, the ground removed from J20-H, 16J1-27, 16J1-22, 16J1-23 and 16J1-28 and the input to J23-31 changed to +5 ±1 Vdc, the assembly shall meet the requirements of 3.2.1.14.1 when measured not less than 3.5 s after changing the input to J23-31.

4.2.2.14.2 B-band/boost. Under the conditions of 4.2.2.14.1, except with the ground applied to 16J1-27, the assembly shall meet the requirements of 3.2.1.14.2.

4.2.2.14.3 C-band. Under the conditions of 4.2.2.14.1, except with the ground removed from J20-e and ground applied to J20-d, the assembly shall meet the requirements of 3.2.1.14.3.

4.2.2.14.4 S-band. Under the conditions of 4.2.2.14.3, except with the ground removed from J20-d and J20-D, surface launch mode selected as specified in 4.1.4.7, the assembly shall meet the requirements of 3.2.1.14.4.

4.2.2.14.5 Pre - ETD inhibit. Under the final conditions of 4.2.2.14.4 with a ground applied to 16J1-28, the assembly shall meet the requirements of 3.2.1.14.5.

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NOTE: It may be necessary, during the performance of 4.2.2.14, to further adjust resistors 16R1 and 16R2 to minimize pitch and yaw drift and assure conformance to 3.2.1.5.

4.2.2.14.6 Alt Bit 2, TTG timing. Under the final conditions of 4.2.2.14.5, except surface launch mode input removed, a ground applied to J20-D, the input to J23-31 replaced with a ground, and the input to J23-27 changed from a ground to +5.0 ±0.5 Vdc, the assembly shall meet the requirements of 3.2.1.14.6.

4.2.2.14.7 Alt Bit 1, TTG timing. Under the final conditions of 4.2.2.14.6, except the ground removed from J20-D, a ground applied to J20-d, and the input to J23-27 changed from +5.0 ±0.5 Vdc to a ground, the assembly shall meet the requirements of 3.2.1.14.7.

4.2.2.15 English bias (5 Vrms 10 Hz input).

4.2.2.15.1 Yaw. With the dc power of 4.1.4.1 applied, the ground removed from J20-R and 5 Vrms at 10 Hz applied to J20-R, the assembly shall meet the requirement of 3.2.1.15.1.

4.2.2.15.2 Pitch. With the final conditions of 4.2.2.15.1 applied, the ground removed from J20-c and 5 Vrms at 10 Hz applied to J20-c, the assembly shall meet the requirement of 3.2.1.15.2.

4.2.2.16 Autopilot outputs.

4.2.2.16.1 Wing lock/latch. With the dc/pick off supply power of 4.1.4.1 applied and with +26.5 Vdc applied to 16J1-30, the assembly shall meet the requirement of 3.2.1.16.1.

4.2.2.16.2 Wing lock/latch (launch). Under the final conditions of 4.2.2.16.1, with the input to J23-31 changed to +5.0 ±0.5 Vdc and J20-H opened, the assembly shall meet the requirement of 3.2.1.16.2.

4.2.2.17 Autopilot BIT.

4.2.2.17.1 Total BIT operation/tune mode. With the dc/pick off supply power of 4.1.4.1 applied, the assembly shall meet the requirement of 3.2.1.17.1.

4.2.2.17.2 1500 Hz. Under the conditions of 4.2.2.17.1 and the amplitude of the 1500 Hz signal at J21-30 reduced to 7.65 Vrms, the assembly shall meet the requirement of 3.2.1.17.2.

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4.2.2.17.3 Total BIT operation/prelaunch mode. With +26.5 ±1.0 Vdc applied to J23-21, full power applied as specified in 4.1.4.6, external hydraulic pressure applied as specified in 4.1.3.1, 0.6 Vrms at 1200 Hz applied between J23-36 and J23-37 (return), the assembly shall meet the requirements of 3.2.1.17.3.

4.2.2.17.4 1200 Hz. Under the final conditions of 4.2.2.17.3, 1.5 Vrms at 1200 Hz applied through a 10 ±1 kohms (10 mW minimum) resistance to 16J1-35, the assembly shall meet the requirement of 3.2.1.17.4.

4.2.2.17.5 Head gyro sense. Under the conditions of 4.2.2.17.4, except with the 1200 Hz signal between J23-36 and J23-37 (return) changed to 0.3 Vrms, the assembly shall meet the requirement of 3.2.1.17.5.

4.2.2.17.6 Wings.

4.2.2.17.6.1 Wings 1 and 3-right. Under the conditions of 4.2.2.17.4, except with a negative dc voltage applied through a 53.6 ±1.1 kohms (10 mW minimum) resistance to 16J1-14 at a level sufficient to obtain a wings 1 and 3 deflection off center of 2.00 ±0.25 deg as specified in 4.1.3.2, the assembly shall meet the requirement of 3.2.1.17.6.1.

4.2.2.17.6.2 Wings 1 and 3-left. Under the conditions of 4.2.2.17.6.1, except with the voltage applied to 16J1-14 changed to a positive dc voltage, the assembly shall meet the requirement of 3.2.1.17.6.2.

4.2.2.17.6.3 Wing 2 up. Under the conditions of 4.2.2.17.6.1, except with the voltage removed from 16J1-14 and applied to 16J1-15, the assembly shall meet the requirement of 3.2.1.17.6.3.

4.2.2.17.6.4 Wing 2 down. Under the conditions of 4.2.2.17.6.3, except with the voltage applied to 16J1-15 changed to positive, the assembly shall meet the requirement of 3.2.1.17.6.4.

4.2.2.17.6.5 Wing 4 up. Under the conditions of 4.2.2.17.6.1, except with the voltage removed from 16J1-14 and applied to 16J1-16, the assembly shall meet the requirement of 3.2.1.17.6.5.

4.2.2.17.6.6 Wing 4 down. Under the conditions of 4.2.2.17.6.5, except with the voltage applied to 16J1-16 changed to positive, the assembly shall meet the requirement of 3.2.1.17.6.6.

4.2.2.18 Autopilot drifts.

4.2.2.18.1 Pitch and yaw drift/B-band. With the assembly oriented to the umbilical up position, a ground applied to J20-L, J20-M, and J20-e; J20-H, 16J1-22, 16J1-23, 16J1-26, 16J1-27 and 16J1-28 ungrounded, full

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power of 4.1.4.6 applied, external hydraulic pressure applied as specified in 4.1.3.1 and then +5 ±1 Vdc applied to J23-31, the assembly shall meet the requirement of 3.2.1.18.1 when measured during a time interval of not less than 10 s beginning not less than 3.5 s after applying +5 ±1 Vdc to J23-31.

NOTE: A ground may be applied to J20-H for not less than 100 ms, to reset the pitch and yaw integrators as necessary during conduct of 4.2.2.18.1.

4.2.2.18.2 Final roll drift/C-band. With the assembly oriented to the umbilical up position, a ground applied to J20-D and J20-d, the grounds removed from J20-H, 16J1-28, 16J1-26 and 16J1-27, full power of 4.1.4.6 applied and external hydraulic pressure applied as specified in 4.1.3.1, the assembly shall meet the requirements of 3.2.1.18.2 after a delay of not less than 3.5 s after +5 ±1 Vdc is applied to J23-31.

4.2.2.19 Accelerometer loops.

4.2.2.19.1 Yaw/1 gravity unit/glide mode-wing 4 up. With the assembly rotated to the wing 4 up position, a ground applied to J20-L, J20-M and J20-e; and with 16J1-22, 16J1-23, 16J1-26, 16J1-27 and 16J1-28 ungrounded, full power of 4.1.4.6 applied and external hydraulic pressure applied as specified in 4.1.3.1, the following steps shall be performed in sequence:

a. +5 ±1 Vdc shall be applied to J23-31 and not less than 3.5 s later, the assembly shall meet the requirements of 3.2.1.19.1a.

b. With the ground removed from J20-H, the assembly shall meet the requirements of 3.2.1.19.1b.

4.2.2.19.2 Yaw/1 gravity unit/glide mode-wing 2 up. Under the conditions of 4.2.2.19.1a, except with the assembly rotated to the wing 2 up position, the input to J23-31 changed to a ground, the assembly shall meet the requirement of 3.2.1.19.2.

4.2.2.19.3 Pitch/1 gravity unit glide mode-wing 3 up. With the assembly rotated to the wing 3 up position, a ground applied to J20-L, J20-M and J20-e; 16J1-22, 16J1-23, 16J1-26, 16J1-27 and 16J1-28 ungrounded, full power of 4.1.4.6 applied and external hydraulic pressure applied as specified in 4.1.3.1, the following steps shall be performed in sequence:

a. +5 ±1 Vdc shall be applied to J23-31 and not less than 3.5 s later, the assembly shall meet the requirements of 3.2.1.19.3a.

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b. With the ground removed from J20-H, the assembly shall meet the requirements of 3.2.1.19.3b.

4.2.2.19.4 Pitch/1 gravity unit/glide mode-wing 1 up. Under the conditions of 4.2.2.19.3, except with the assembly rotated to the wing 1 up position, the input to J23-31 grounded, the assembly shall meet the requirements of 3.2.1.19.4.

4.2.2.20 Wing centering. With the grounds removed from 16J1-22, 16J1-23 and 16J1-26, the dc power of 4.1.4.1 applied, with external hydraulic pressure applied as specified in 4.1.3.1, the assembly shall meet the requirement of 3.2.1.20.

4.2.2.21 Mode commands/surface to surface. With 5 Vrms at 28 ±6 Hz applied to J20-g, the assembly shall meet the requirement of 3.2.1.21.

4.2.2.21.1 Doppler acquisition mode select. With 1 Vrms at 115 Hz applied to J20-P, the assembly shall meet the requirement of 3.2.1.21.1.

4.2.2.22 Resistance measurement.

4.2.2.22.1 Squib resistance. (All inputs and loads may be removed.) With the condition at all contacts of 4.1.4.5 changed to an open, the resistance between J20-J and J20-A (ground) and between J21-29 and J21-10 shall be measured using an igniter circuit tester (Alinco model 101-5CFG or equivalent) which applies less than 5 mA to the connector contact under test (J20-J), the assembly shall meet the requirement of 3.2.1.22.1.

WARNING

Excessive squib current (>5 mA) may ignite the squibs.

4.2.2.22.2 Triggering device. The assembly shall meet the requirement of 3.2.1.22.2.

4.2.2.22.3 Video inject shield. The assembly shall meet the requirement of 3.2.1.22.3.

4.2.2.22.4 Continuity. The assembly shall meet the requirements of 3.2.1.22.4.

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5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1), preservation and packaging shall be Level A or Level C as defined in FED-STD-102.

5.1.1 Level A. Each assembly shall be individually protected and unit packaged in accordance with MIL-P-116, Method III.

5.1.2 Level C. The assembly shall be packaged to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Packages shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packing shall be Level A or C as defined in FED-STD-102.

5.2.1 Level A. The assembly, packaged as specified in 5.1.1, shall be packed in wood-cleated fiberboard; nailed wood; wirebound, corrugated, or solid fiberboard; or wood-cleated plywood (overseas type) boxes.

5.2.2 Level C. Assemblies packaged in accordance with 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.3.2 Normal marking. Unit packages and shipping containers shall be in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The control section is intended for use in the AIM/RIM-7M guided missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Government-furnished property list (see 3.1.1).

c. Minimum standards for working environments, if different than 3.3.2.1.

d. Responsibility for inspection and inspection facilities, if different than 4.1.

e. Levels of preservation, packaging and packing (see 5.1 and 5.2).

f. Special markings (see 5.3.1).

6.3 Subject term (keyword) listing.

AC to DC converter
Accumulator
Autopilot
Craft gyro assembly
Flow valve assembly
Hydraulic control section
Missile tune converter

6.4 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity
Navy-AS
(Project 5865-N063)

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